



IT3020 - Database Systems (DBS)

B.Sc. (Hons) in IT

Laboratory Worksheet 01

This practical is aimed at refreshing your previous knowledge of relational SQL and familiarity with Oracle. It would be a good idea to write down the SQL before coming to the practical and also bring along a softcopy if you want to save time at the practical session.

1. Use the following relational database tables for recording the information shown below. Primary keys are underlined>. (Assume each client is given a client number for unique reference.)

The screenshot shows the FreeSQL web interface. In the SQL Worksheet, the following SQL statement is entered and executed:

```
1 CREATE TABLE client (  
2   client CHAR(3) PRIMARY KEY,  
3   name VARCHAR(12),  
4   address VARCHAR(30)  
5 );
```

The output shows: "Table CLIENT created. Elapsed: 00:00:00.016". The Navigator on the left shows the schema structure with CLIENT under My Schema.

The screenshot shows the FreeSQL web interface. In the SQL Worksheet, the following SQL statement is entered and executed:

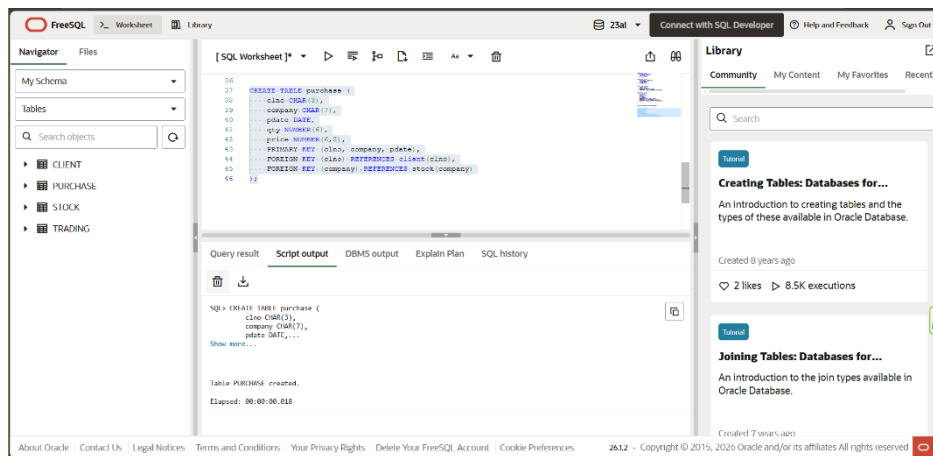
```
7 CREATE TABLE stock (  
8   company CHAR(7) PRIMARY KEY,  
9   price NUMBER(4,2),  
10  dividend NUMBER(4,2),  
11  eps NUMBER(4,2)  
12 );
```

The output shows: "Table STOCK created. Elapsed: 00:00:00.019". The Navigator on the left shows the schema structure with STOCK under My Schema.

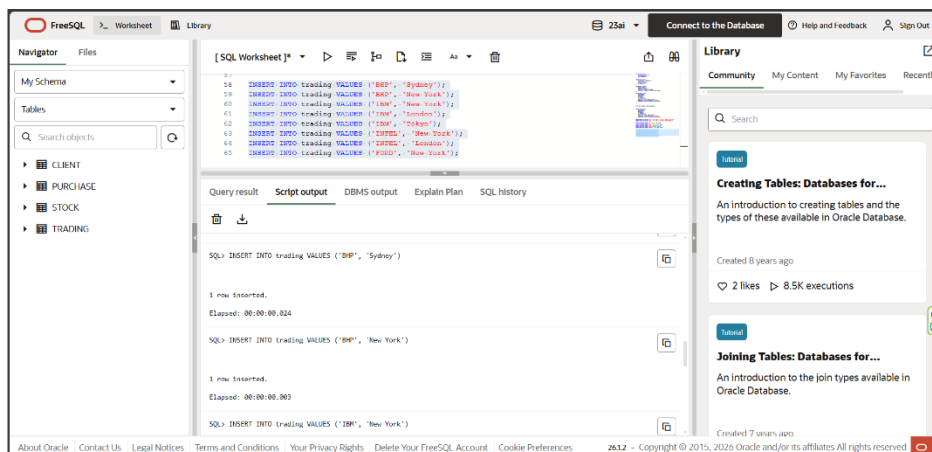
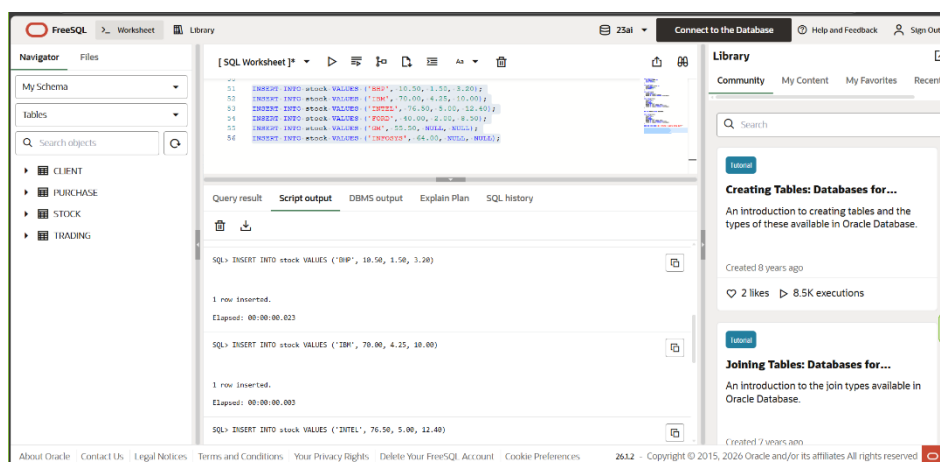
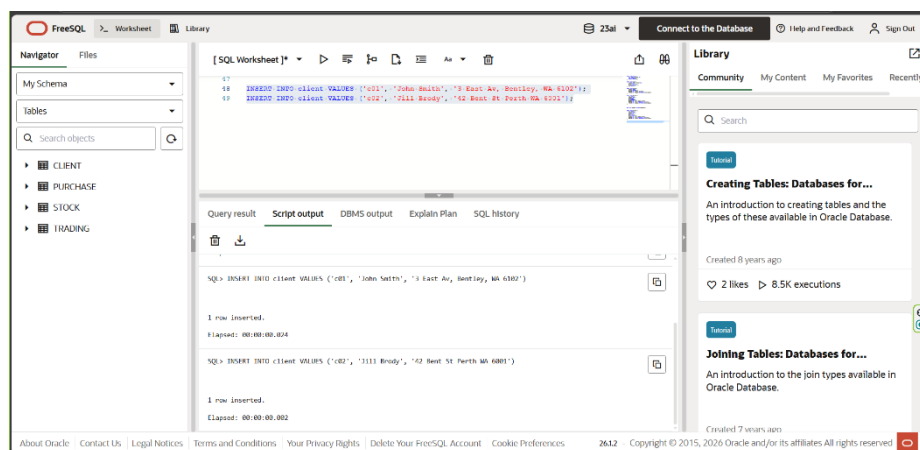
The screenshot shows the FreeSQL web interface. In the SQL Worksheet, the following SQL statement is entered and executed:

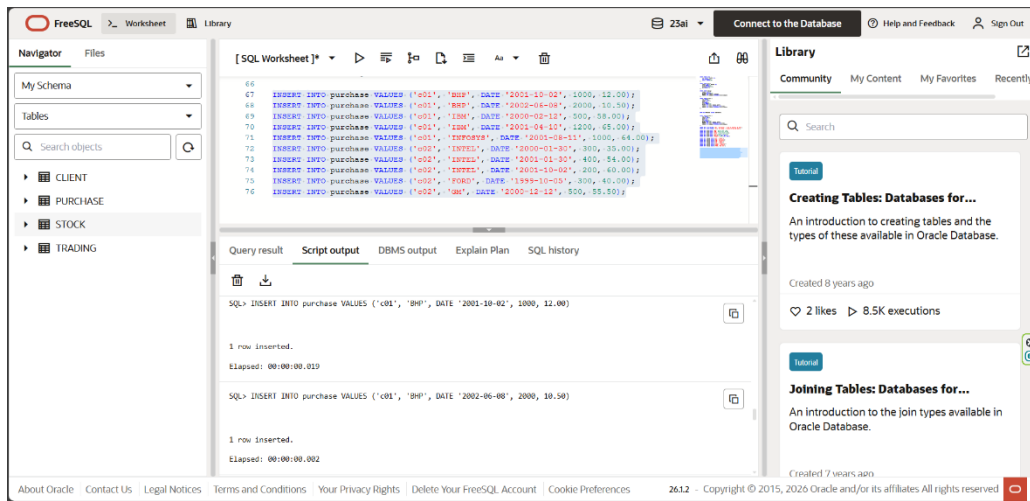
```
13 CREATE TABLE trading (  
14   company CHAR(7),  
15   exchange VARCHAR(12),  
16   PRIMARY KEY (company, exchange),  
17   FOREIGN KEY (company) REFERENCES stock(company)  
18 );
```

The output shows: "Table TRADING created. Elapsed: 00:00:00.020". The Navigator on the left shows the schema structure with CLIENT, STOCK, and TRADING under My Schema.



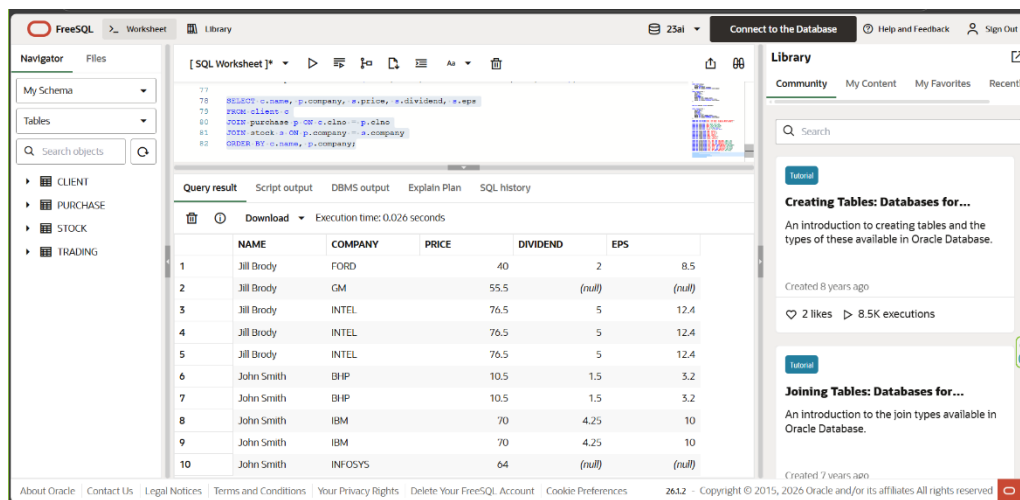
2. Create the relational tables in Oracle with suitable primary and foreign keys, and insert the sample data given below.



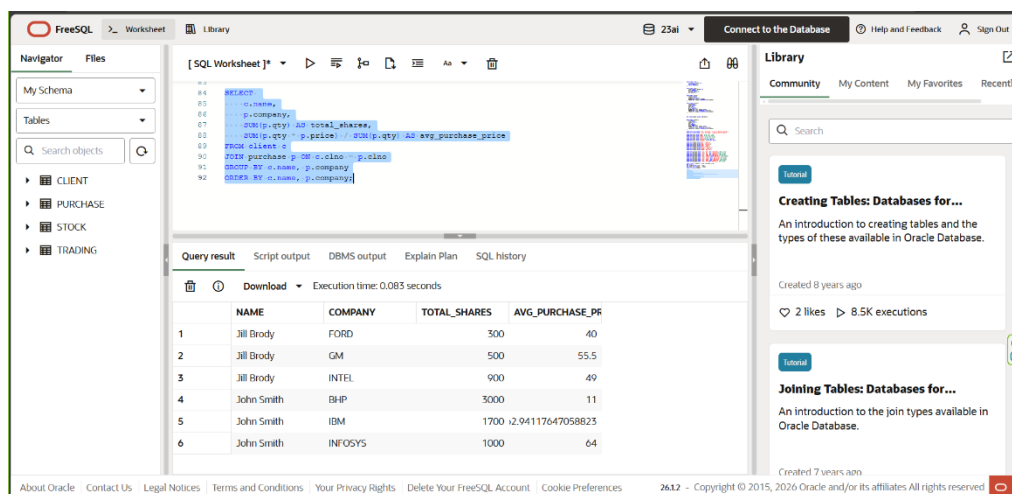


3. Answer the queries given below, using the tables in your database.

(a) For each client, get the client's name, and the list of the client's investments with stock name, current price, last dividend and earnings per share.



(b) Get the list of all clients and their share investments, showing the client name, and for each stock held by the client, the name of the stock, total number of shares held, and the average purchase price paid by the client for the stock. Average price is the total purchase value paid by a client for a given stock (value=qty*price) divided by the total quantity held by the client.



- (c) For each stock traded in New York, find the quantity held by each client, and its current value (value=qty*price). Display stock (company) name, client name, number of shares held, and the current value of the shares.

The screenshot shows the FreeSQL interface with a SQL query in the editor and its results in the 'Query result' tab. The query is as follows:

```

SELECT
  t.company,
  c.name,
  SUM(p.qty) AS total_shares,
  SUM(p.qty * p.price) AS current_value
FROM trading t
JOIN purchase p ON t.company = p.company
JOIN client c ON p.cno = c.cno
JOIN stock s ON p.cno = s.cno
WHERE s.exchange = 'New York'
GROUP BY t.company, c.name
ORDER BY t.company, c.name

```

The results table shows the following data:

	COMPANY	NAME	TOTAL_SHARES	CURRENT_VALUE
1	BHP	John Smith	3000	31500
2	FORD	Jill Brody	300	12000
3	IBM	John Smith	1700	119000
4	INTEL	Jill Brody	900	68850

- (d) Find the total purchase value of investments for all clients. Display client name and total purchase value of the client's portfolio.

The screenshot shows the FreeSQL interface with a SQL query in the editor and its results in the 'Query result' tab. The query is as follows:

```

SELECT
  c.name,
  SUM(p.qty * p.price) AS total_purchase_value
FROM client c
JOIN purchase p ON c.cno = p.cno
GROUP BY c.name
ORDER BY c.name

```

The results table shows the following data:

	NAME	TOTAL_PURCHASE
1	Jill Brody	83850
2	John Smith	204000

- (e) For each client, list the book profit (or loss) on the total share investment. Book profit is the total value of all stocks based on the current prices less the total amount paid for purchasing them.

The screenshot shows the FreeSQL interface with a SQL query in the editor and its results in the 'Query result' tab. The query is as follows:

```

SELECT
  c.name,
  SUM(p.qty * s.price) - SUM(p.qty * p.price) AS book_profit_loss
FROM client c
JOIN purchase p ON c.cno = p.cno
JOIN stock s ON p.cno = s.cno
GROUP BY c.name
ORDER BY c.name

```

The results table shows the following data:

	NAME	BOOK_PROFIT_LOS
1	Jill Brody	24750
2	John Smith	10500